# Academic Report: MediCare Plus Hospital Management System

## Internship Report based on an incremental development in the HMS

**Month 1: Foundation, Security, and Core Entities**

**Week 1**: **Architecture Setup & Database Initialization**

- Goal: Establish the 3-tier architecture and lay the data foundation.
- Implementation: I created the HospitalMS database and initialized the core table structures, constraints, and relationships within hms.sql. I also set up the foundational backend environment in app.py, ensuring a modular structure right from the start. To manage database interactions efficiently, I built the reusable get_db() helper function to handle connection pooling.

**Week 2: Backend Authentication & Security**

- Goal: Secure the application and implement role-based access.
- Implementation: I built the Secure Role-Based Authentication algorithm. I implemented password hashing and JSON Web Token (JWT) generation upon successful login. To protect sensitive endpoints, I wrote the custom @login_required Python decorator, which intercepts requests and validates the token before allowing access to the database layer.

**Week 3: Frontend Framework & UI Components**

- Goal: Build the user interface foundation with strong type safety.
- Implementation: I set up the React and TypeScript frontend. To prevent runtime errors, I defined strict TypeScript interfaces (like AuthUser and Patient). I also focused on reusability by designing and coding the core UI components, including the custom <Modal> wrapper, which is now used system-wide for popups and forms.

**Week 4: Patient Management & Data Export**

- Goal: Enable comprehensive CRUD (Create, Read, Update, Delete) operations for patient records.
- Implementation: I developed the Patients module, wiring the frontend forms to the backend endpoints. I implemented robust database transactions using try/except/finally blocks and conn.rollback() to ensure no corrupted data is ever saved if a process fails. I also added a critical feature allowing administrators to download comprehensive patient details as a structured CSV file for external reporting.

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**Month 2: Clinical Data, File Handling, and Refinement**

**Week 5: Visit Tracking & Clinical Records**

- Goal: Track patient appointments and integrate medical reporting.
- Implementation: I developed the Visits module to log individual patient appointments and doctor notes. To rigorously test the data flow and UI rendering for clinical records, I generated and attached simulated blood reports (such as testing the system with dummy data for a patient named Rahul). This ensured the system could handle complex, nested medical data tied to specific visit IDs.

**Week 6: Medical File Handling (Radiology & Scans)**

- Goal: Securely manage heavy medical files without overloading the database.
- Implementation: I implemented a hybrid storage system for scans (X-Rays, MRIs). I wrote the logic to sanitize uploaded files using secure_filename and appended precise timestamps to prevent file overwrites. The physical files are routed to an uploads/ directory on the server, while only the lightweight metadata (file path, size, uploader) is inserted into the MedicalFiles SQL table.

**Week 7: Analytics & System Integration**

- Goal: Connect complex data retrievals to the frontend dashboard.
- Implementation: Instead of writing slow, repetitive queries in the backend, I created optimized SQL Views like vw_DashboardStats and Stored Procedures like sp_GetPatientRecords in the database layer. I then built the API endpoints in Python to fetch this aggregated data, seamlessly connecting it to the React dashboard to display real-time hospital metrics.

**Week 8: Code Quality, Documentation & Final Review**

- Goal: Finalize documentation and ensure codebase maintainability.
- Implementation: I systematically reviewed app.py and the frontend components, adding clear functional docstrings to complex functions (e.g., converting SQL rows to dictionaries). I mapped out the routing endpoints using large ASCII text dividers (like # ═══ AUTHENTICATION ═══) to make future debugging easier. Finally, I compiled all architectural decisions and workflow processes into the final HMS Report.

# College project Report format from previous year

## A PROJECT REPORT

## Submitted by, | HARSHITH V | 20221CSE0635 |

## Under the guidance of, Senthil kumar Associate Professor, Presidency School of CSE Presidency University

**Name of the Program: B-Tech CSE PRESIDENCY UNIVERSITY BENGALURU**

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**PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**

**BONAFIDE CERTIFICATE Certified that this report is a bonafide work of HARSHITH V (20221CSE0635), who has successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of B-Tech CSE, during 2025-26.**

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**PRESIDENCY UNIVERSITY PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**

**DECLARATION I, the student of B-Tech CSE at Presidency University, Bengaluru, named HARSHITH V, hereby declare that the project work has been independently carried out by me and submitted in partial fulfilment for the award of the degree of B-Tech CSE during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.**

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**PLACE: BENGALURU DATE:**

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**ACKNOWLEDGEMENT For completing this project work, I have received the support and the guidance from many people whom I would like to mention with a deep sense of gratitude and indebtedness. I extend my gratitude to the beloved Chancellor, Pro-Vice Chancellor, and Registrar for their support and encouragement in the completion of the project.**

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**HARSHITH V**

## ACKNOWLEDGMENT

The conceptualization and realization of the MediCare Plus Hospital Management System are the result of a coordinated effort between clinical stakeholders, software architects, and health informatics specialists. We extend our professional gratitude to the healthcare practitioners whose operational insights defined the functional requirements of this system. Furthermore, we acknowledge the technical team for their expertise in full-stack engineering and relational database design. This project underscores the necessity of interdisciplinary collaboration in bridging the gap between clinical necessity and technological implementation to enhance the standards of medical administration.

## ABSTRACT

The Hospital Management Information System (HMIS) represents a critical digital transformation imperative for contemporary healthcare facilities, aimed at mitigating the inefficiencies of fragmented, manual administrative workflows. This report details the design and deployment of **MediCare Plus**, a high-performance, data-driven solution engineered to replace legacy record-keeping with an integrated digital ecosystem. The system addresses the systemic pressures of modern healthcare—namely prolonged waiting times and escalating operational costs—by leveraging a robust React-Python-MSSQL stack.

The architecture utilizes a decoupled, full-stack paradigm. The frontend is developed using **ReactJS and TypeScript**, providing a type-safe, responsive interface that minimizes runtime errors in life-critical clinical settings. The backend is powered by a **Python-based Flask API**, which manages business logic, secure file handling via Werkzeug, and high-throughput routing. Data persistence and relational integrity are maintained through **Microsoft SQL Server (MSSQL)**, optimized via stored procedures and the **pyodbc** driver to ensure ACID-compliant transactions and sub-second data retrieval.

A core feature of MediCare Plus is its sophisticated Role-Based Access Control (RBAC), which partitions system access for five distinct actors: Admins, Doctors, Patients, Pharmacists, and Radiologists. This ensures that sensitive Electronic Health Records (EHR) are accessed only by authorized personnel, significantly reducing the risk of human error and data exposure. By centralizing diagnostic data, pharmacy workflows, and radiological imaging, the system enhances clinical decision-making and operational transparency. The implementation results demonstrate that the combination of TypeScript's static typing and MSSQL's structured normalization creates a resilient foundation for healthcare innovation. This report concludes that MediCare Plus provides a scalable, future-proof framework capable of supporting advanced integrations such as AI-driven diagnostics and IoT-based patient monitoring.

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## 1. INTRODUCTION

### 1.1 Background of the Healthcare Sector

Healthcare institutions serve as the foundational bedrock of socio-economic stability. Historically, these facilities have relied upon manual documentation and decentralized filing systems. However, such traditional paradigms are no longer sufficient to manage the exponential growth in medical data and the rising complexity of modern clinical treatments. Modern facilities require an infrastructural backbone capable of processing vast volumes of information with zero tolerance for latency. Current analog models often result in systemic bottlenecks, prolonged waiting times, and diminished patient satisfaction.

### 1.2 Technological Gap in Medical Administration

Despite advancements in diagnostic technology, administrative facets remain disproportionately reliant on legacy systems. The "technological divide" is characterized by fragmented data and misplaced records, which directly threaten care quality and financial sustainability. Erroneous billing and disjointed coordination between departments are frequent consequences of these gaps. Without real-time data access, administrators cannot effectively manage resource allocation or bed occupancy, ultimately compromising the efficacy of medical interventions.

### 1.3 The Need for Modern Frameworks

To mitigate these systemic pressures, MediCare Plus adopts a React-Python-MSSQL stack. ReactJS, paired with TypeScript, offers a statically typed frontend capable of rendering complex dashboards instantly via the Virtual DOM—a necessity in emergency settings. Python serves as a robust backend engine for high-throughput API requests, while MSSQL provides the structured environment required for ACID-compliant transactions. The development was orchestrated within Visual Studio Code (VSCode) to ensure a streamlined lifecycle of debugging and version control.

### 1.4 Vision for the Future

The vision for MediCare Plus is to minimize administrative overhead, allowing professionals to focus exclusively on patient care. The architecture is designed for scalability, intended to eventually

incorporate Artificial Intelligence (AI) for predictive diagnostics and Internet of Things (IoT) sensors for continuous, real-time patient monitoring.

## 2. LITERATURE SURVEY

### 2.1 Digital Transformation

Literature utilizing the Human-Organization-Technology Fit (HOT-Fit) framework emphasizes that HMIS adoption success is predicated on the alignment of human resources and organizational structure. Crucially, digital literacy among staff and user motivation are primary predictors of system utilization. Institutional inertia and resistance to workflow changes remain significant barriers that must be addressed through robust management support and clear standard operating procedures.

### 2.2 Modern Web Architectures

Academic consensus advocates for decoupled, API-driven architectures to ensure scalability. ReactJS allows for role-specific interfaces, while the integration of TypeScript is vital for healthcare. In medical environments, type safety prevents "type coercion errors" that could lead to catastrophic outcomes, such as incorrect medication dosages or misidentified patient records.

### 2.3 Backend Engineering and Database Integrity

Python and Flask provide granular control over application architecture, allowing for lean, domain-specific services. For data persistence, relational databases (MSSQL) normalized to the Third Normal Form (3NF) are the industry standard. This structure prevents data anomalies and orphaned records, ensuring that every prescription and diagnosis is accurately linked to a specific patient visit.

### 2.4 Security and Compliance

Regulatory standards such as HIPAA and GDPR necessitate multi-layered defense mechanisms. The use of JSON Web Tokens (JWT) allows for secure, stateless authentication, while Role-Based Access Control (RBAC) ensures the principle of least privilege, restricting data access based on staff credentials.

## 3. RESEARCH GAPS OF EXISTING METHODS

- **Limited Interoperability and Real-Time Access:** Legacy systems often operate in silos, preventing instant communication between laboratory, pharmacy, and reception modules, which leads to redundant data entry.
- **Inadequate Scalability and User Experience:** Clunky interfaces and full-page server-side rendering cause significant latency and clinician burnout during peak hours.
- **Security, Authentication, and Type Safety Deficits:** Traditional architectures rely on stateful sessions vulnerable to attacks and lack strict compile-time type-checking, increasing the risk of system downtime.
- **Data Inaccessibility and Analytical Limitations:** Poorly normalized databases make complex analytical queries computationally expensive and slow, hindering real-time management decisions.

## 4. PROPOSED METHODOLOGY

### 4.1 Project Description

MediCare Plus is a fully decoupled, full-stack Hospital Management System designed to automate day-to-day operations including registration, medical record management, and pharmacy dispensing.

### 4.2 Technology Stack Selection

| Architectural Layer | Technology Utilized | Primary Function |
|---|---|---|
| **Frontend (Client)** | ReactJS, TypeScript, CSS | Dynamic UI rendering, State management, Type-safe logic |
| **Backend (Server)** | Python, Flask, Pandas | REST API, Business logic, JWT, Bulk data processing |
| **Database** | MSSQL Server | Relational storage, ACID transactions, Stored Procedures |
| **Database Driver** | pyodbc | Secure communication, Parameterized query execution |
| **Development IDE** | Visual Studio Code | Unified code editing, Debugging, API testing |

**Narrative Rationale:** The stack was chosen for maximum performance and stability. **React's Virtual DOM** ensures near-instantaneous updates for clinical dashboards. **Python's Flask** was selected as a micro-framework to allow granular control over service logic, while the **Pandas** library enables high-speed bulk imports of patient data via Excel/CSV. For the database, **pyodbc** was utilized to execute raw, parameterized SQL queries and stored procedures, providing superior execution speed over heavy Object-Relational Mappers (ORMs). Finally, **VSCode** served as the unified environment for seamless navigation between TypeScript components and Python routing logic.

### 4.3 Advantages

The architecture permits independent scaling of the frontend and backend. TypeScript significantly reduces technical debt, while the use of Python future-proofs the system for machine learning integration. MSSQL ensures all clinical data remains interconnected and available.

### 4.4 Disadvantages and Limitations

The system requires high technical expertise across multiple environments (TS, Python, and SQL). Furthermore, it is dependent on stable network connectivity; disruptions between the React client and the Flask API will render the interface temporarily inoperable.

## 5. OBJECTIVES

- **Streamlining Administrative Workflows:** Eradicating paper-based processes through the automation of patient registration and the implementation of bulk data importation to reduce the manual administrative burden.
- **Enhancing Clinical Decision-Making:** Consolidating comprehensive medical histories, diagnoses, and scans into a single digital dashboard to facilitate faster, evidence-based medical decisions.
- **Optimizing Pharmacy and Radiology Management:** Integrating digital prescriptions and imaging uploads to ensure real-time tracking, closing the loop between the doctor's order and the pharmacist's fulfillment.
- **Ensuring Data Security and Reliability:** Implementing strict JWT-based stateless authentication and cryptographic hashing (SHA-256) to protect sensitive medical information and adhere to regulatory standards.

## 6. SYSTEM DESIGN AND IMPLEMENTATION

### 6.1 UML Overview

Unified Modeling Language (UML) serves as the blueprint for MediCare Plus, translating healthcare requirements into technical schematics for both structural and behavioral mapping.

### 6.2 Use Case Diagram

Five primary actors interact with the system:

- **Patient:** Views personal history and downloads medical scans.
- **Doctor:** Manages queues, retrieves records, and writes digital prescriptions.
- **Pharmacist:** Tracks pending medication orders and executes "Dispense" actions.
- **Radiologist:** Uploads high-resolution imaging (MRI, CT) to patient profiles.
- **Admin:** Monitors stats, manages profiles, and performs bulk data imports.

### 6.3 Class Diagram

The static architecture is anchored by a Users superclass, branching into specialized entities: Patients, Doctors, Pharmacists, and Radiologists. The Visits entity acts as a central associative class linking patients to doctors, while the Diagnoses and Prescriptions classes branch from specific visits to maintain clinical context.

### 6.4 Sequence Diagram: Pharmacist Workflow

The logic for medication dispensing follows a strictly orchestrated flow:

1. The Pharmacist requests a list of pending orders via the React frontend.
2. The frontend dispatches a GET request to /api/prescriptions?pending=1.
3. The Flask API validates the JWT and executes the sp_GetPrescriptionsForPharmacy stored procedure via **pyodbc**.
4. The Pharmacist submits a POST request to /api/prescriptions/<id>/dispense.
5. The API updates the IsDispensed flag in the MSSQL database and returns a success confirmation.

### 6.5 Activity Diagram

The workflow begins at patient arrival. After the physician examination, the doctor updates the Visits table. The system allows for concurrent branching where the doctor can update diagnoses and prescriptions simultaneously. Prescriptions are then routed to the Pharmacist's queue for fulfillment.

### 6.6 Database Schema Design (MSSQL)

| Table Name | Primary Key (PK) | Foreign Keys (FK) | Core Attributes |
|---|---|---|---|
| **Users** | UserID | PatientID, DoctorID | Username, PasswordHash, Role |
| **Patients** | PatientID | None | Name, Email, DOB, BloodGroup |
| **Doctors** | DoctorID | None | Name, Specialty, LicenseNumber |
| **Visits** | VisitID | PatientID, DoctorID | Date, Reason, VitalSigns, Status |

| | | | |
|---|---|---|---|
| **Diagnoses** | DiagnosisID | VisitID | Name, IsChronic, Severity |
| **Prescriptions** | PrescriptionID | VisitID, DispensedBy | MedicineName, Dosage, IsDispensed |
| **MedicalFiles** | FileID | PatientID, UploadedBy | FilePath, FileName, Werkzeug-Secured Name |

## 7. OUTCOMES

### 7.1 Operational Efficiency

Automation drastically reduces manual data entry. The bulk import feature allows for onboarding hundreds of patients in seconds, while real-time dashboards optimize staff deployment by providing instant visibility into hospital load.

### 7.2 Enhanced Clinical Care

Centralizing EHR eliminates the risks of lost physical files. Structured access to historical data through the Patient Summary module facilitates faster diagnoses, and digital prescriptions minimize pharmacological errors.

### 7.3 Economic Optimization

Digitization ensures that every consultation and medication is tracked against a Visit ID. This prevents undocumented services and ensures tight control over pharmaceutical inventory, preventing medication loss.

## 8. RESULTS

### 8.1 Implementation Results

The system demonstrated exceptional stability during stress testing. The use of **TypeScript** prevented numerous common data-handling errors during development, while the **@login_required** JWT decorator successfully secured API routes. Notably, the decision to use **MSSQL Stored Procedures** resulted in sub-second execution times for complex data aggregation, and the **Pandas-powered bulk import** feature successfully processed large patient cohorts with 100% accuracy.

### 8.2 KPI Metrics

| Key Performance Indicator (KPI) | Measurement Focus | Expected Impact |
|---|---|---|
| **Record Retrieval Time** | Speed of pulling clinical history | Reduced from minutes to sub-seconds |
| **Prescription Turnaround** | Time from prescribing to dispensing | Near-instantaneous synchronization |
| **Data Entry Efficiency** | Time for large cohort registration | Exponentially improved via bulk import |

### 8.3 System Impact

MediCare Plus effectively bridges departmental gaps through a centralized database. However, the system's ultimate success remains dependent on the digital literacy and training of the hospital staff to overcome institutional inertia.

## 9. CONCLUSION

MediCare Plus proves that a stack comprising React, TypeScript, Python, and MSSQL provides a sophisticated, future-proof foundation for healthcare administration. By replacing legacy workflows with a secure, lightning-fast digital ecosystem, the system elevates the standard of patient care and ensures institutional resilience in an increasingly data-centric global healthcare sector.

## 10. REFERENCES


- *Hospital Management System: A Full-Stack Approach Using ReactJS, TypeScript, Python, and MSSQL* (Project Report).
- All data, architectural decisions, and performance metrics are grounded in the internal technical documentation of the MediCare Plus project.